Q26 Deeper Project Suggestions

1. **Rank sweep on the stability–plasticity curve:** extend this module's NumPy demo to sweep LoRA rank $r \in \{1, 2, 4, 8, 16, 64\}$ on a two-task sequence; plot task-A retention against task-B gain and identify the knee. Compare against full fine-tuning with replay at several replay ratios. Where does the rank constraint beat data mixing, and why?
2. **Retrieve-vs-tune ablation on a real corpus:** pick a domain (e.g. a course's own lecture notes or an arXiv slice). Build (i) a frozen base model prompted zero-shot, (ii) the same model + a retrieval index, and (iii) the same model LoRA-tuned on the corpus. Evaluate on factoid questions vs. format/style tasks. Test the module's rule: does retrieval win facts and tuning win behavior on your stack?
3. **Anchored online updates:** implement a streaming adaptation loop where a small-r LoRA is refreshed on incoming minibatches against a pinned reference model, with an explicit KL anchor. Characterize drift, entropy collapse, and forgetting as functions of anchor strength; find the setting where naive online SGD visibly breaks and the anchored version does not.
4. **Pipeline-order experiment:** on a domain-adaptation task, run SFT→DPO and DPO→SFT orderings with matched compute and data. Measure domain behavior quality, preference-metric scores, and regression on general benchmarks. Does the "preference last" ordering claim survive your setup?